Fossils: An introduction to fossil preservation Answers

Definition of a fossil:

A fossil represents part of a once living plant or animal or it has been formed by the actions of a once living plant or animal. A fossil or trace fossil also has to be prehistoric – from a time before any historical written records.

Some types of plants and animals are more likely to become fossilised than others e.g. an organism that dies on the land surface is liable to be scavenged, dismembered and decayed before there is any chance of burial and preservation. An organism which lives on the sea floor where there is rapid sedimentation stand a much better chance of burial and removal from scavengers and agents of decay. So, in general, marine organisms have a better preservation potential than terrestrial organisms.

Decide which of these organisms has the greater chance of being preserved as a fossil and write down your reasons: -

a) a cow or a seal?
   Although both a cow and a seal are mammals with robust skeletons the seal has the higher preservation potential due to the fact that it spends most of its life in the ocean, unlike the cow!
b) A jellyfish or an oyster?
   The oyster because it has a mineralised shell composed of calcium carbonate, in the form of the mineral calcite.
c) Plankton or pollen?

   Pollen is produced by seed bearing plants (common from the Mesozoic onwards) and so are terrestrial, however, they are very small and very resistant to chemical change and produced in vast numbers, so preservation potential is high – part of specialist subject called palynology see this web site for more details - https://www.ucl.ac.uk/GeolSci/micropal/spore.html

   Plankton include both plants and animals and are organisms that live in water and are carried by drifting within the water column. Plankton found as fossils can include the diatoms, radiolarian, foraminifera, nanno planktons like coccolithophores

   https://www.ucl.ac.uk/GeolSci/micropal/welcome.html

   Both plankton and pollen have a high preservation potential, but the marine realm is much larger than the terrestrial, so overall I would have to go for plankton.

Material:

There are nine fossil samples

Binocular microscopes, hand lenses, grain size cards and dilute HCl

Task:

Take only one specimen to work on at a time

Use the table to record how each fossil has been preserved

The state of preservation of fossil material and its relationship to the enclosing sediments can provide much evidence to help with the interpretation of past environments and this is a skill that all geologists should aim for. Whilst you are focussing on the preservation style of the fossils you should also make a note of the observable characteristics of any sedimentary rock surrounding the fossil – note colour, grain size, mineral composition, bedding, sorting if you can observe them.

Note: You are not expected to be able to identify the fossils – they have been labelled for you – you are observing & describing the style of preservation

Sample Number	Fossil name	Type of preservation	Rock description	How did this fossil form
LL…….(Museum sample)	Various insects – including ants & mosquitos	Unaltered soft parts	Amber	Amber = sticky tree resin, traps insects, hardens, volatiles escape
F25	Plant root - *Stigmaria*	Internal mould	Medium sandstone	
F26	Gastropods & bivalves	External moulds and internal moulds	Limestone "Portland Stone"	All original shell material dissolved leaving voids in the rock

F27	Plant stem - *Sigillaria*	Carbonisation	Black siltstone/mudstone	Volatile compounds lost – carbon left as thin black film
F30	Ammonite *Liparoceras*	Unaltered hard parts – original shell		Very little change except removal of unstable organic matter
F31	Borings in a fossil coral – Boring = *Gastrochoenolites*	Trace Fossil	Limestone	Boring in the hard substrate of the coral skeleton
F32	Bivalve	Internal mould “osses ead”	Fine-medium sandstone	“Steinkern” – original shell material dissolved away
F33	Ammonite – *Psiloceras planorbis*	Unaltered hard parts – colour from the original mother of pearl	Grey coloured siltstone	Very little change
F34	Ammonite	Replacement with calcite	Dark grey mudstone	Pyritisation – original shell

		and pyrite crystals		has been replaced with Fe S

You can choose from the following modes of preservation: -

Unaltered soft parts; Altered soft parts; Unaltered hard parts; Carbonisation; Permineralisation; Recrystalisation; Replacement; External mould; Internal mould; Cast or Trace Fossil.